

KETCHIKAN, AK, USA

WMO: 703950

Lat: 55.357N

Lon: 131.712W

Elev: 24

StdP: 101.03

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25325

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-9.6	-6.9	-18.4	0.7	-5.8	-15.2	1.0	-4.8	12.5	4.6	11.4	5.3	1.9	290	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.0	22.5	16.1	20.8	15.3	19.2	14.5	16.9	21.2	16.0	19.5	15.2	18.1	4.5	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.8	17.6	14.5	10.3	16.8	13.9	9.9	16.2	47.3	21.2	44.6	19.6	42.5	18.1	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.2	8.2	-13.2	25.9	3.4	1.9	-15.7	27.3	-17.7	28.4	-19.7	29.4	-22.2	30.8		
(5)			-13.9	18.0	3.2	1.0	-16.2	18.7	-18.1	19.3	-19.8	19.8	-22.2	20.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.7	2.1	2.1	3.2	6.2	9.8	12.7	14.7	14.8	11.9	7.9	4.1	2.1
	DBStd	5.63	4.34	3.28	3.10	2.29	2.71	2.41	2.30	2.14	2.08	2.44	3.29	3.77
	HDD10.0	1333	245	222	210	117	37	2	0	0	7	72	177	246
	HDD18.3	3901	503	455	468	365	264	170	117	112	192	324	427	504
	CDD10.0	483	0	0	0	2	31	83	146	149	64	6	1	0
	CDD18.3	9	0	0	0	0	0	2	4	3	0	0	0	0
	CDH23.3	32	0	0	0	0	1	8	17	6	1	0	0	0
	CDH26.7	3	0	0	0	0	0	1	1	1	0	0	0	0
(14) Wind	WSAvg	3.6	3.7	3.6	3.7	3.8	3.6	3.6	3.5	3.4	3.5	3.6	3.6	3.6
(15) Precipitation	PrecAvg	4099	359	322	285	290	232	199	191	267	340	599	433	390
	PrecMax	4712	794	602	631	572	534	364	446	540	685	1076	1092	609
	PrecMin	2519	95	21	70	52	78	45	24	24	45	326	71	169
	PrecStd	555	163	158	143	137	114	80	114	140	163	198	219	134
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	10.2	12.9	16.8	21.5	24.5	25.6	24.0	19.7	14.7	11.4	10.0
		MCWB	7.3	6.6	7.7	8.7	12.2	16.0	17.9	17.1	14.4	11.2	8.5	7.0
	2%	DB	8.8	8.6	9.7	13.5	18.5	21.1	22.7	22.1	17.4	13.0	10.2	8.7
		MCWB	7.2	6.1	6.6	8.5	12.0	14.7	16.5	16.4	13.4	10.4	8.9	7.1
	5%	DB	7.8	7.4	8.3	11.5	16.5	19.0	20.6	20.4	16.2	12.2	9.3	7.9
		MCWB	6.6	5.8	6.1	7.4	11.2	13.7	15.6	15.7	13.1	10.1	8.1	6.6
	10%	DB	7.0	6.5	7.4	10.1	14.6	17.1	18.8	18.8	15.0	11.3	8.4	7.0
		MCWB	6.0	5.2	5.6	6.9	10.3	13.0	14.7	15.0	12.7	9.7	7.3	6.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.1	8.6	10.5	13.9	16.6	18.3	18.0	15.7	12.8	10.4	8.5
		MCDB	9.6	8.8	10.6	13.6	18.8	22.5	24.5	22.1	17.8	13.7	10.8	9.0
	2%	WB	7.8	6.9	7.7	9.1	12.7	15.2	17.0	17.0	14.6	11.7	9.1	7.8
		MCDB	8.5	7.7	9.2	11.7	17.0	19.5	21.8	20.8	16.3	12.3	9.8	8.4
	5%	WB	6.9	6.2	6.8	8.2	11.7	14.3	15.9	16.2	13.8	10.8	8.3	6.9
		MCDB	7.5	7.1	8.0	10.6	15.1	18.0	19.7	19.5	15.3	11.4	9.1	7.6
	10%	WB	6.3	5.4	5.9	7.6	10.8	13.3	15.1	15.4	13.2	10.1	7.6	6.0
		MCDB	7.0	6.5	7.1	9.7	13.5	16.5	18.1	18.0	14.6	10.9	8.4	6.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	5.0	5.8	6.7	7.1	6.3	5.8	6.0	5.5	5.0	4.2	4.1
		MCDBR	4.2	5.2	7.1	9.5	10.9	10.5	9.5	9.1	7.5	5.6	4.2	4.1
		MCWBR	3.4	3.8	4.7	5.3	5.5	5.0	4.6	4.4	4.2	3.9	3.5	3.2
	5% WB	MCDBR	3.8	4.4	5.9	7.4	9.0	9.1	8.4	8.0	5.7	4.0	3.8	3.8
		MCWBR	3.4	3.7	4.4	4.6	4.9	4.5	4.3	4.0	3.6	3.5	3.5	3.4
(40) Clear-Sky Solar Irradiance	taub		0.297	0.327	0.333	0.366	0.367	0.369	0.376	0.359	0.343	0.333	0.307	0.292
	taud		2.363	2.356	2.429	2.346	2.418	2.451	2.466	2.538	2.563	2.488	2.343	2.261
	Ebn at Noon		639	748	842	857	877	877	863	858	827	740	610	535
	Edh at Noon		52	74	87	108	107	105	101	89	76	65	52	45
(44) All-Sky Solar Radiation	RadAvg		0.47	1.25	2.20	3.65	4.57	4.74	4.23	3.70	2.41	1.32	0.59	0.36
	RadStd		0.06	0.26	0.38	0.29	0.48	0.38	0.56	0.37	0.33	0.21	0.10	0.07

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	-190	+84	N/A
(47) Regional (16 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60	N/A

Nomenclature: See separate page